

Supplementary Material — A Pre-Registered Ledger of Feature Families for Single-Energy CT Stone-Composition Prediction: What Works, and What Does Not

S1. Rationale

The clinically actionable question in single-energy non-contrast CT (NCCT) stone imaging is whether stone composition can be estimated well enough, from data already collected at presentation, to guide definitive management. The dominant decisions are uric-acid vs. non-uric-acid (dissolvable vs. not) and, among calcium stones, **calcium oxalate (CaOx) vs. calcium phosphate (CaP)** — a distinction with metabolic and preventive consequences. Struvite is not a modeling target: infection stones are a clinical diagnosis (alkaline, turbid, malodorous urine at pH 7–8.5; dysuria, flank pain, recurrent UTI, frequently impending sepsis) that any emergency clinician makes without a model.

This supplement reports, in one place, **every feature family we tested** — the hypothesis and mechanistic rationale that motivated each, the extraction/derivation method, the evaluation, the result, and the disposition — with the successes grouped alongside the failures. Negative results are reported deliberately and in full: publishing only positive findings leaves others to repeatedly expend resources on avenues that do not bear fruit. Our aim is to delineate the boundary of what single-energy NCCT plus routinely available data can and cannot resolve, so that subsequent work is directed at the informative gaps (Section S4).

Unless stated otherwise, evaluation is **repeated stratified patient-grouped 5-fold cross-validation** with **patient-clustered bootstrap 95% CIs**; the calcium-subtype tests use a dedicated CaOx-vs-CaP binary head (positive class = CaP) with the deltas expressed against the **HU + routine-labs** baseline (AUROC 0.710). Cohort: 319 laboratory-analyzed stones from 223 patients (CaOx 107, CaP 144, UA 36, struvite 27, other 5). The ledger spans two endpoints — **composition** (which stone type; Table S1) and **multicomponent detection** (whether a stone is mixed at all; Table S2, §S5).

S2. Master ledger (Table S1)

#	Domain	Feature family	Hypothesis / rationale	Key result	Verdict
Retained					
R1	Model	Lean UA model (stone peak/mean HU; urine pH; CO ₂ /Cl/anion gap; BUN, Cr, Ca, glucose; age, sex)	UA is acid-driven and low-HU; routine data should identify it	AUROC 0.781 (0.677–0.867) , NPV 0.97, Brier 0.086; positive net benefit across 10–50% thresholds	Deployed
R2	Model	CaOx-vs-CaP head, HU + routine labs	Acid–base chemistry (pH, RTA pattern, Ca) separates the calcium subtypes better than HU alone	HU 0.665 → HU+labs 0.710 (0.637–0.778) ; beats 5-class CaP OvR (0.674)	Deployed (cascade + high-sensitivity CaP screen)
R3	Imaging	Nephrocalcinosis (parenchymal calcific foci)	dRTA / hyperparathyroidism / MSK cause parenchymal Ca deposition and CaP stones	Biology clean (CaP frac 10× CaOx) but head Δ +0.013 (CI crosses 0); outlier-driven	Radiologic flag, not a model feature
Rejected — imaging					
F1	Imaging	Hb-anchored HU calibration (air + aortic-blood anchors)	Normalize scanner HU offset to sharpen composition-HU mapping	No same-composition variance reduction (scanner offset ~10 HU ≪ ~220 HU biological spread)	Rejected
F2	Imaging	Self-supervised 3D CNN (frozen-encoder deep features)	Learned volumetric features may capture texture beyond handcrafted HU	Underperformed; deep features did not cluster by composition; degraded fusion	Rejected
F3	Imaging	HU-distribution-shape radiomics (skew, kurtosis, entropy, GMM modes, Shapiro)	Multi-component / zoned stones have distinctive HU histograms	Reduced cross-validated AUC	Rejected
F4	Cohort	Multi-stone N-lift (volume ↔ mass rank matching, 131 → 319)	More labeled stones per patient → power	Neutral (kept for N; no lift)	Rejected (as a lift)
F5	Imaging	Stone density gradient (core/periphery HU ratio)	CaOx retains a dense core; CaP tapers	CaOx 2.86 vs CaP 2.80 (gap 0.06 vs SD 1.2); head Δ −0.013; SHAP 13/17	Rejected
F6	Labs	Normal-AG (hyperchloremic) acidosis composite + hyperchloremia index	Explicit dRTA/GI-loss signature the tree may not synthesize	Δ −0.000 (CI −0.009,+0.008); redundant with raw CO ₂ /Cl/AG	Rejected
F7	Labs	eGFR (CKD-EPI 2021)	CKD shifts mineral metabolism toward CaP	Δ −0.009; SHAP rank 7 yet OOF-flat (redundant with Cr/age/sex)	Rejected
F8	Imaging	Opportunistic vertebral BMD (L1–L3 trabecular HU)	Hypercalciuria/dRTA ↔ low BMD ↔ calcium stones	Full-cohort +0.028 but complete-case crosses 0; flat by-composition (187 vs 194); 35% missing (FOV)	Rejected
F9	Imaging	Hepatic steatosis (liver–spleen HU)	Gut dysbiosis raises enteric oxalate (CaOx) and hepatic fat → fatty liver > in CaOx	Steatosis CaOx 13% vs CaP 17% (p=0.71, reversed); Δ −0.006; signal is UA (already captured)	Rejected

#	Domain	Feature family	Hypothesis / rationale	Key result	Verdict
F10	Imaging	Sarcopenia / muscle quality (paraspinal+psoas HU, volume)	Body composition indexes metabolic phenotype	$\Delta -0.023$ (P 0.03, worse); muscle-volume SHAP rank 5 while hurting (body-size confound)	Rejected
F11	Imaging	Aortic calcification (Agatston-style, denoised)	$\text{Ca} \times \text{PO}_4$ / CKD-MBD mineral axis (no serum PO_4 /PTH in data) $\rightarrow$ CaP	Age-adjusted coef -0.10 (CI $-0.41, +0.21$); $\text{CaOx} \geq \text{CaP}$ (330 vs 249); head $\Delta +0.001$	Rejected
F12	Imaging	Bladder wall fraction / volume	Chronic cystitis/obstruction $\rightarrow$ struvite	Bladder-only $\Delta +0.017$ (P 0.79); distension-confounded	Rejected
F13	Imaging	Prostate calcification	Chronic prostatitis $\rightarrow$ struvite	SHAP rank 1/22 but a sex proxy (79% M / 1% F); non- significant; lowers 5-class struvite AUC 0.686 $\rightarrow$ 0.626	Rejected (leak)

S3. Detail by family

A. Retained models (R1–R2) and the one flag (R3)

The **lean UA model** (R1) is the primary clinical product: at a high-sensitivity operating point it rules out uric acid with NPV 0.97 and shows positive decision-curve net benefit across the plausible threshold range. The **CaOx-vs-CaP head** (R2) is the only feature *addition* that improved the calcium-subtype decision: adding the routine metabolic panel to stone HU raised AUROC from 0.665 to 0.710 — the discriminating information for the calcium split is **chemical, not morphological** (Main Figs 6–7). **Nephrocalcinosis** (R3) is the sole imaging feature with clean, correct biology (mean parenchymal-calcification fraction $\sim 10\times$ higher in CaP than CaOx; prevalence CaOx 21% $\rightarrow$ CaP 33%); it is retained as a **deterministic radiologic flag** that prompts a CaP/dRTA/hyperparathyroidism work-up when substantial foci are auto-detected, because as a learned feature it is carried by a handful of high-burden cases and does not survive outlier removal ($\Delta +0.013 \rightarrow +0.004$).

B. Rejected imaging features (F1–F5, F8–F13)

These span three mechanistic bets. **HU fidelity** (F1 calibration, F2 CNN, F3 shape radiomics, F5 density gradient): none beat handcrafted peak/mean HU — the composition signal in attenuation is low-dimensional and adding HU-derived complexity overfits at this N. **Bone/mineral-metabolism surrogates** (F8 BMD, F11 aortic calcification): both index axes (bone loss, vascular calcium) that are dominated by **age** — already modeled — and, for vascular calcium, index the hypercalciuria/CaOx phenotype as much as the phosphate/CaP one, netting to null. **Systemic/metabolic surrogates** (F9 hepatic steatosis, F10 sarcopenia, F12–F13 bladder/prostate): each re-encodes the metabolic-syndrome/urine-pH axis (F9's steatosis signal is real but belongs to UA, already captured) or, worse, a confound — F13 prostate calcification is a **sex indicator** (present in 79% of males, 1% of females) that the model exploits, ranked first by SHAP yet non-significant and harmful to its intended target class.

C. Rejected lab/derived features (F6–F7)

The routine chemistry is fully exploited by the raw analytes: an explicit normal-anion-gap-acidosis composite (F6) and CKD-EPI eGFR (F7) are deterministic functions of values already in the model (CO₂/Cl/anion gap; creatinine/age/sex) and add nothing out-of-fold, despite SHAP attending to eGFR (rank 7).

S4. Methodological through-line and interpretation

Three recurring findings organize the ledger:

1. **Feature importance is not predictive value.** Repeatedly, a feature drew high SHAP attention while adding nothing (eGFR, rank 7) or actively hurting (muscle volume, rank 5; prostate calcification, rank 1) out-of-fold. Adjudication was therefore on grouped-CV Δ AUROC with patient-clustered bootstrap CIs, never on importance. Figure S1 shows every family's Δ against the HU+labs baseline: **all confidence intervals cross zero.**
2. **Apparent gains require verification.** The one nominally significant full-cohort result (nephrocalcinosis + BMD, +0.038) dissolved under complete-case analysis (removing the 35% of scans without lumbar spine in the field of view: +0.024, CI crossing zero) and outlier removal. The bladder/prostate "gain" (+0.039) was a sex-proxy leak. Every candidate was subjected to complete-case, age-adjusted, and/or outlier-robustness checks before disposition.
3. **The information boundary.** Across thirteen rejected families spanning HU fidelity, mineral-metabolism surrogates, systemic-metabolic surrogates, and derived chemistry, single-energy composition prediction is **saturated by stone attenuation and urine/acid–base chemistry.** The residual CaOx-vs-CaP ceiling (~0.71) is not a modeling limitation to be engineered away with additional CT-derived or routine-lab features; it reflects the absence of the **stone-specific analytes** that define the two supersaturation states — most directly **urinary oxalate** (the swing driver of CaOx supersaturation) and its modifiers (magnesium, citrate), which are not present in routine data and are not inferable from single-energy attenuation.

S5. A second endpoint — multicomponent detection (an exhaustive negative)

Recasting the problem around management decisions (main text §3.7) raises a distinct, coarser question grounded in stone pathophysiology: most stones are **multicomponent** (68% here), laminating a nidus with a shell of different composition — so **can any routine-data channel detect that a stone is mixed, without resolving what it is mixed of?** This is a lower bar than composition subtyping (heterogeneity, not identity) with better-balanced classes, and it was tested exhaustively. Every feature family of the study — plus a dedicated **peak-deconvolution** family added for this question (per-stone 1- vs 2-Gaussian mixture fits: peak separation, Sarle's bimodality coefficient, skewness, and the 2-vs-1 BIC evidence) — was evaluated against the multicomponent label under the same patient-grouped CV with clustered bootstrap.

Table S2. Multicomponent detection by channel (multicomponent = second-component fraction $\geq 10\%$; AUROC, patient-clustered bootstrap 95% CI; Figure S4).

Channel	n feat	AUROC (95% CI)	Verdict
Stone HU (base)	5	0.552 (0.480–0.625)	null
HU-shape radiomics	10	0.553 (0.482–0.619)	null
Density gradient	1	0.426 (0.353–0.496)	below chance
Hb-calibrated HU	3	0.422 (0.356–0.491)	below chance
CNN deep features	64	0.480 (0.413–0.545)	null
Labs / urine	10	0.484 (0.415–0.549)	null
Derived acid–base + eGFR	3	0.433 (0.360–0.507)	null
Demographics	2	0.435 (0.367–0.504)	null
Nephrocalcinosis	4	0.357 (0.290–0.428)	below chance
Vertebral BMD	2	0.459 (0.390–0.527)	null
Hepatic steatosis	3	0.569 (0.488–0.649)	null
Sarcopenia	2	0.481 (0.407–0.559)	null
Aortic calcification	3	0.427 (0.360–0.494)	below chance
Bladder / prostate	3	0.414 (0.345–0.485)	below chance
Peak deconvolution	5	0.527 (0.455–0.598)	null (0.375 at $\geq 25\%$)
All combined	115	0.626 (0.557–0.695)	UA-purity confound

No channel detects multicomponent structure. Every individual family's interval crosses chance, several below it. The single apparent gain — the 115-feature combined model at **0.626 (0.557–0.695)** — is a **uric-acid-purity confound**, not multicomponent detection: uric-acid stones are **39% multicomponent vs. 72% for non-UA**, so predicting "mixed" is largely predicting "not-UA," which the model already does (Node A). Two orthogonal controls expose it: restricting to calcium stones (removing the UA axis) drops the combined model to **0.549 (non-significant)**, and a clean $\geq 25\%$ mixedness threshold drops it to **0.491 (chance)**.

Peak deconvolution — the most direct attempt, actively fitting two overlapping Gaussians and measuring their separation and skewness (two unequal overlapping peaks sum to a skewed composite) — was itself at chance (0.53, size-robust set excluding the voxel-count-confounded BIC), *below* chance at the clean threshold (0.375), added nothing to the HU-shape moments (0.526 $\rightarrow$ 0.541), and was matched by a **stone-volume-only baseline** — what little varied was stone size, not bimodality. At clinical slice thickness (≈ 3 mm) partial-volume averaging erases the two-peak structure. The lamination is histologically real but not resolvable on single-energy clinical CT: **the multicomponent architecture that characterizes most stones is undetectable by any channel a standard ED visit provides** — a reinforcement, at a coarser and better-powered endpoint, of the composition-subtyping boundary in Table S1.

S6. Figures

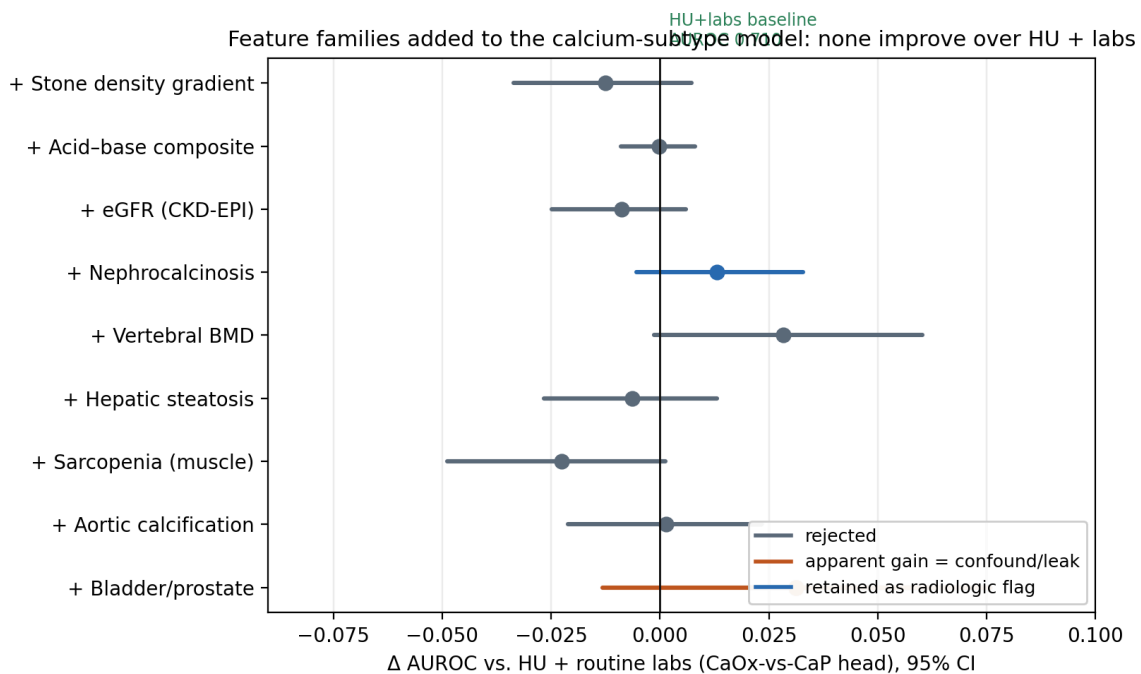


Figure S1. Forest plot of Δ AUROC for each feature family added to the HU+labs CaOx-vs-CaP head (patient-clustered bootstrap 95% CI). All intervals cross zero; the only genuine improvement was adding the routine metabolic panel to stone HU (baseline, AUROC 0.710). Grey = rejected; orange = apparent gain traced to a confound/leak (bladder/prostate = sex proxy); blue = retained as a radiologic flag (nephrocalcinosis).

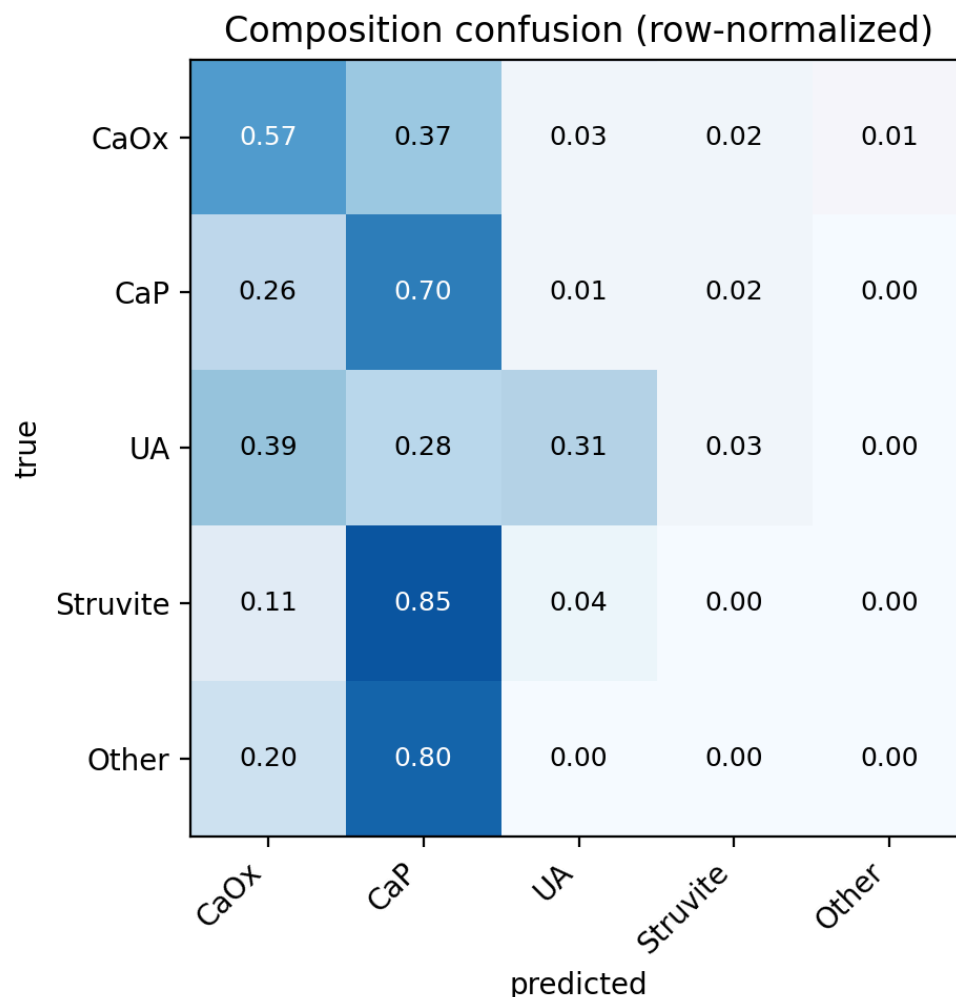


Figure S2. Five-class composition confusion matrix (row-normalized), patient-level cross-validation.

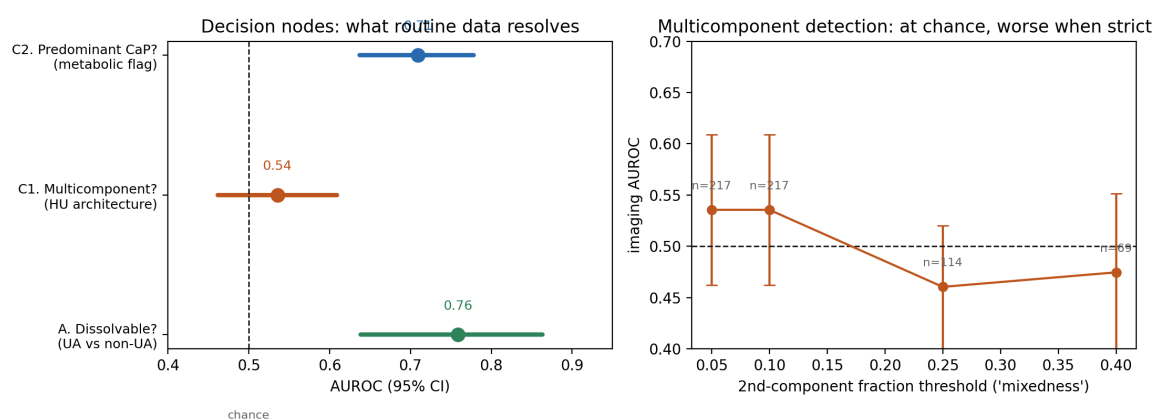


Figure S3. Decision-node performance (left) and the multicomponent threshold sweep (right). Routine data resolves the dissolvable decision (Node A, uric acid vs. non-UA) and a modest calcium-phosphate metabolic flag (Node C2), but multicomponent detection (Node C1) sits at chance and falls below it as the mixedness definition tightens.

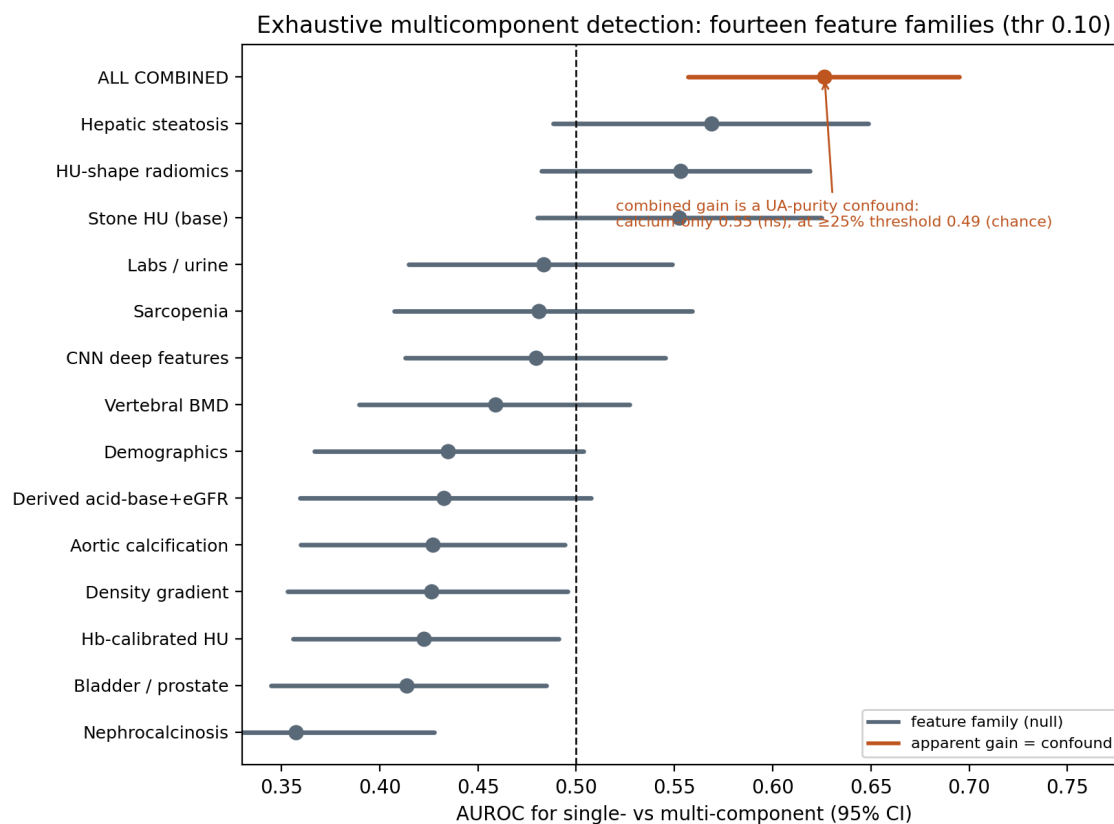


Figure S4. Exhaustive multicomponent detection — every feature family plus peak deconvolution against the single- vs. multi-component endpoint (thr 0.10; patient-clustered bootstrap 95% CI). All families cross or sit below chance; the lone combined "gain" (orange) is a uric-acid-purity confound (see Table S2). - Main-text figures cross-referenced: Fig 1 UA ROC; Fig 2 decision-curve; Fig 3 calibration; Fig 4 UA SHAP; Fig 5 density-gradient per-class AUC; Fig 6 CaOx-vs-CaP head ROC by feature set; Fig 7 CaP-screen operating point.